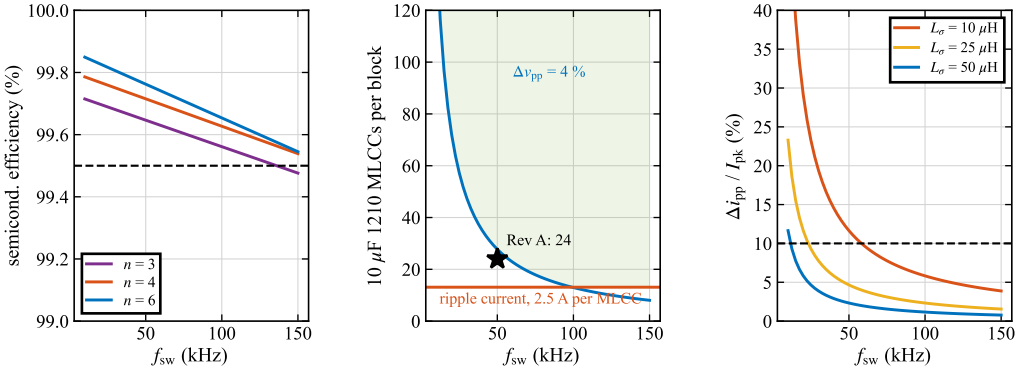
semicond. efficiency (%)
100.0
99.8
99.6
99.4
99.2
99.0
50
100
150
$f_{sw}$ (kHz)
$n = 3$
$n = 4$
$n = 6$
(a)
10 $\mu$F 1210 MLCCs per block
120
100
80
60
40
20
0
$\Delta v_{pp} = 4$ %
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ripple current, 2.5 A per MLCC
50
100
150
$f_{sw}$ (kHz)
(b)
$\Delta i_{pp}$ / $I_{pk}$ (%)
40
35
30
25
20
15
10
5
0
$L_\sigma = 10$ $\mu$H
$L_\sigma = 25$ $\mu$H
$L_\sigma = 50$ $\mu$H
50
100
150
$f_{sw}$ (kHz)
(c)